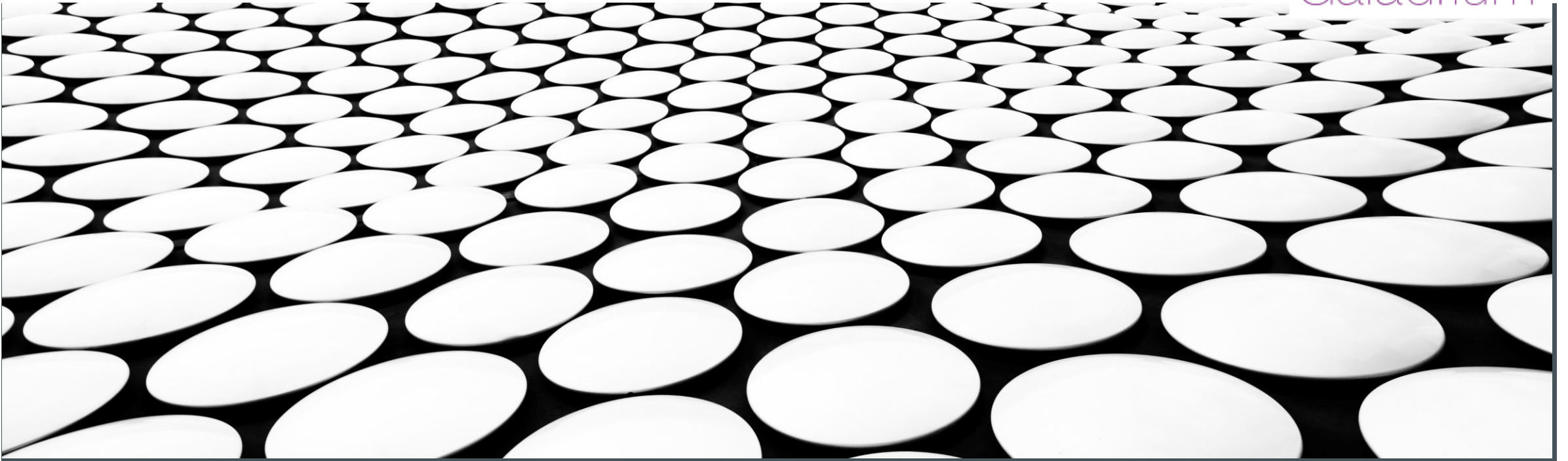

ATTECH TRAINING: AMHS/SWIM GATEWAY USE CASES

EXAMPLES OF THE USE OF AMHS/SWIM GATEWAYS

Galadrium



Italy ENAV

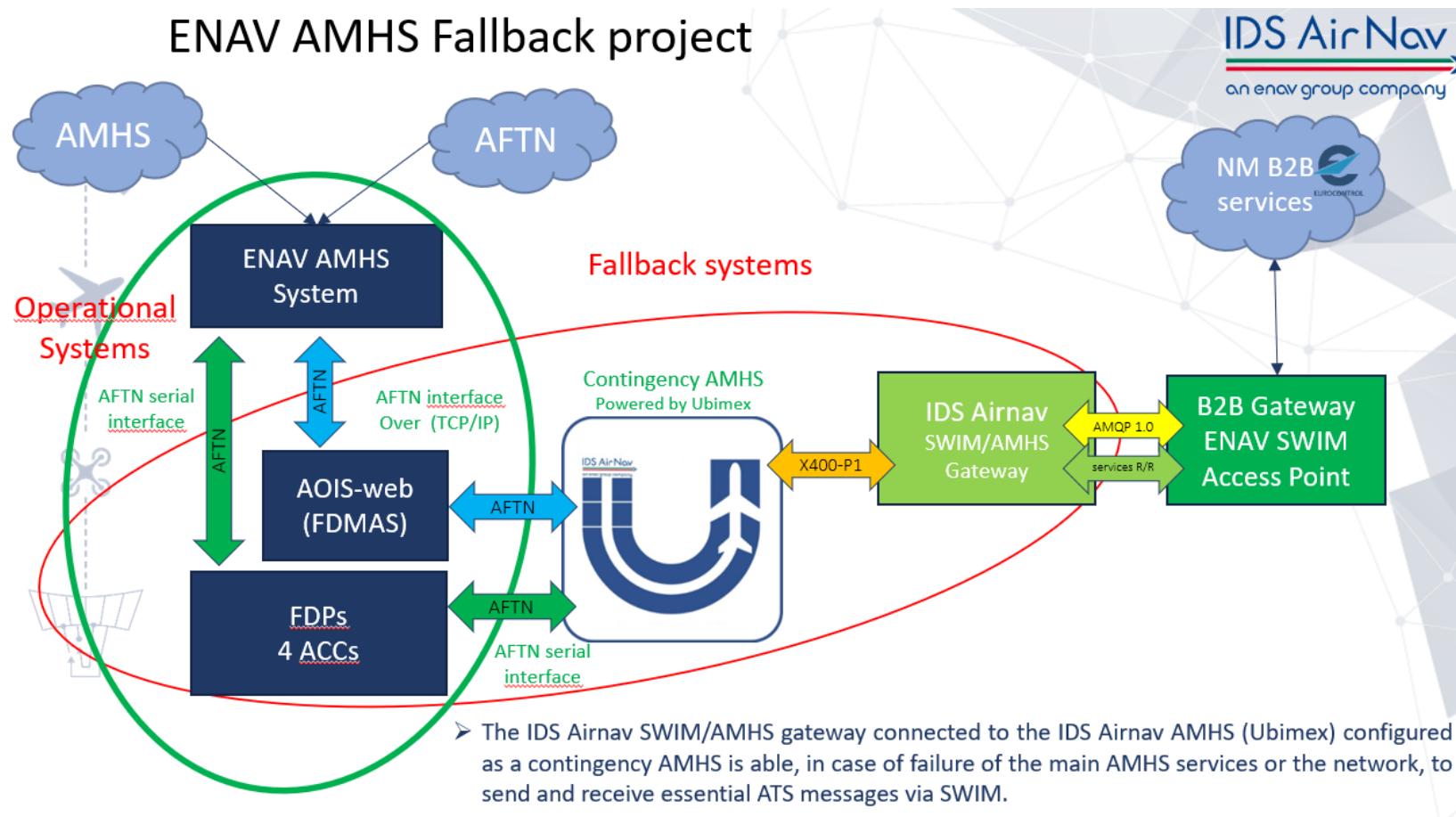


ENAV ITALY

- ENAV is the Italian ANSP
- They use an AMHS system developed by IDS
- IDS is now a company that is part of ENAV
- IDS have developed an ASG
- Their AMHS system has a contingency system to connect to the Eurocontrol NM using the ASG

6) Use cases of the AMHS/SWIM Gw

- Use case #1 (By ENAV, COM Center back-up functionality)



FUM and DPI



FUM AND DPI MESSAGES

- FUM stands for Flight Update Messages
- DPI stands for Departure Planning Information
- They are messages required in the Airport Collaborative Decision Making (A-CDM) systems
- They allow airports to send real-time departure information and receive arrival updates (FUMs)
- They enhance efficiency, reduce delays and improve overall flow management



DISCONTINUATION OF FUM AND DPI MESSAGES ON AFTN/AMHS

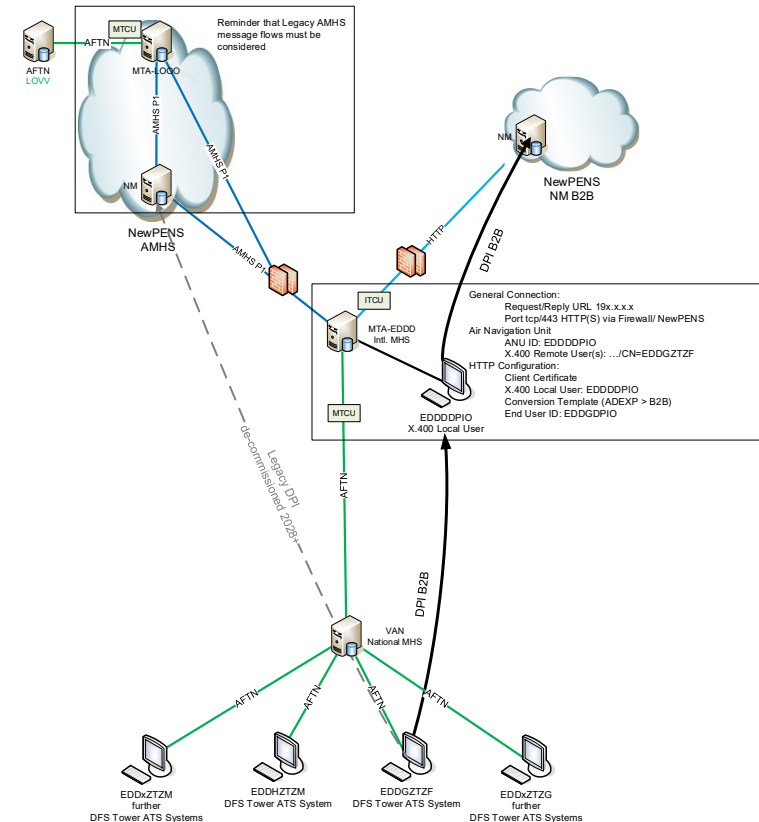
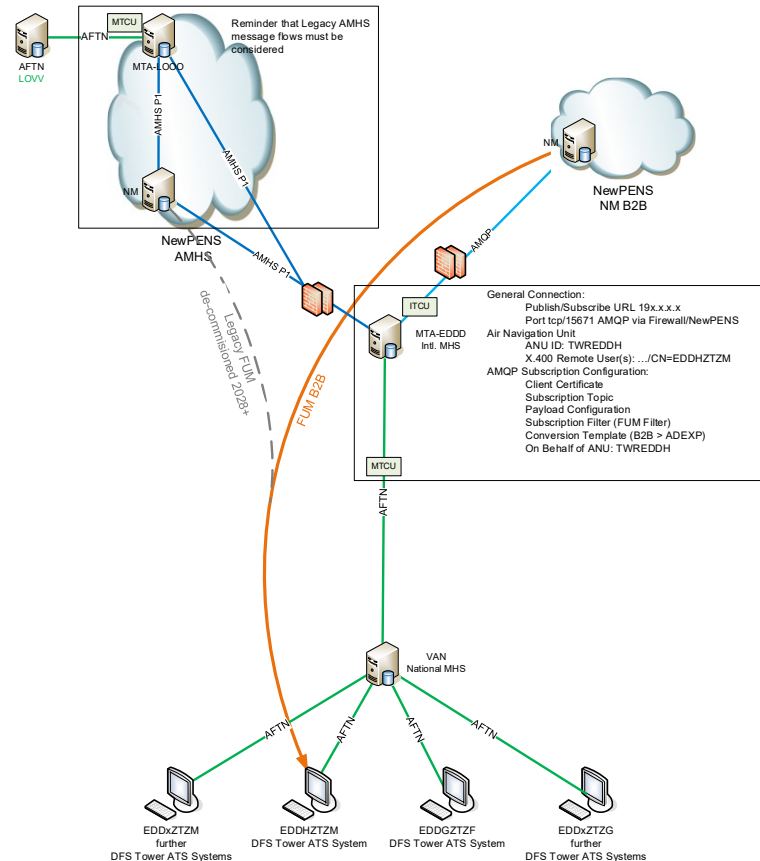
- The Eurocontrol Network Manager (NM) announced recently the discontinuation of delivery of FUM and DPI messages over AMHS and AFTN at the end of 2027
- While the DFS ATS Tower Systems are being re-developed, the systems need to continue to receive the messages via AMHS

DFS USE CASE FOR FUM & DPI

- DFS is the German ANSP
- They presented an Information Paper to the AST-TF PG informing of the decision to use the ASG
- They will use ASG to convert between SWIM AMQP message to AMHS and vice-versa
- ENAIRE (Spain) also identified the use of the ASG, but in their case they decided to use a direct NM B2B connection
- NATS (UK) was also looking the issue of using an ASG, or leaving the airports to connect themselves
- Representatives of Poland and Denmark also notified of ongoing activities to offer a solution

6) Use cases of the AMHS/SWIM Gw

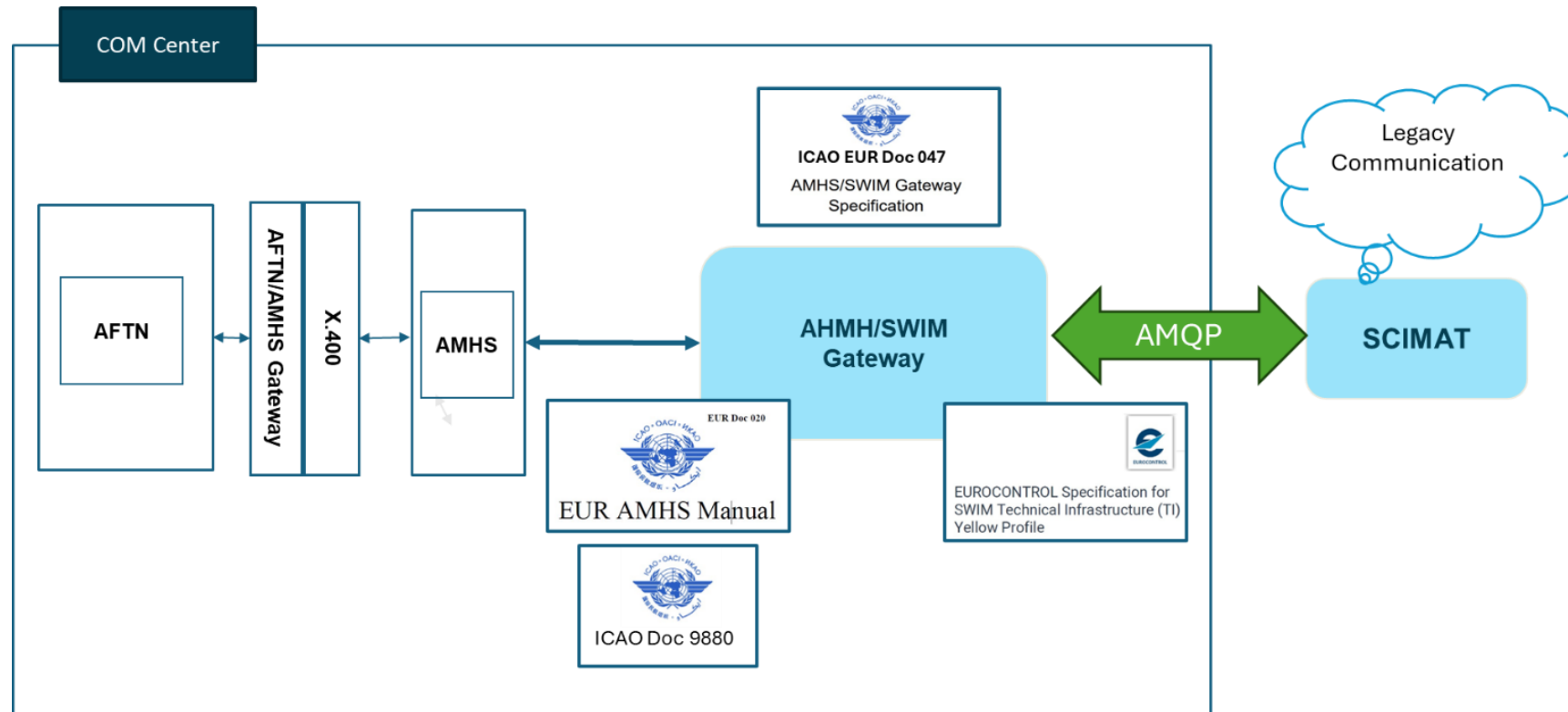
- Use case #2 (By DFS, FUM & DPI messages)



ENNAIRE SCIMAT

6) Use cases of the AMHS/SWIM Gw

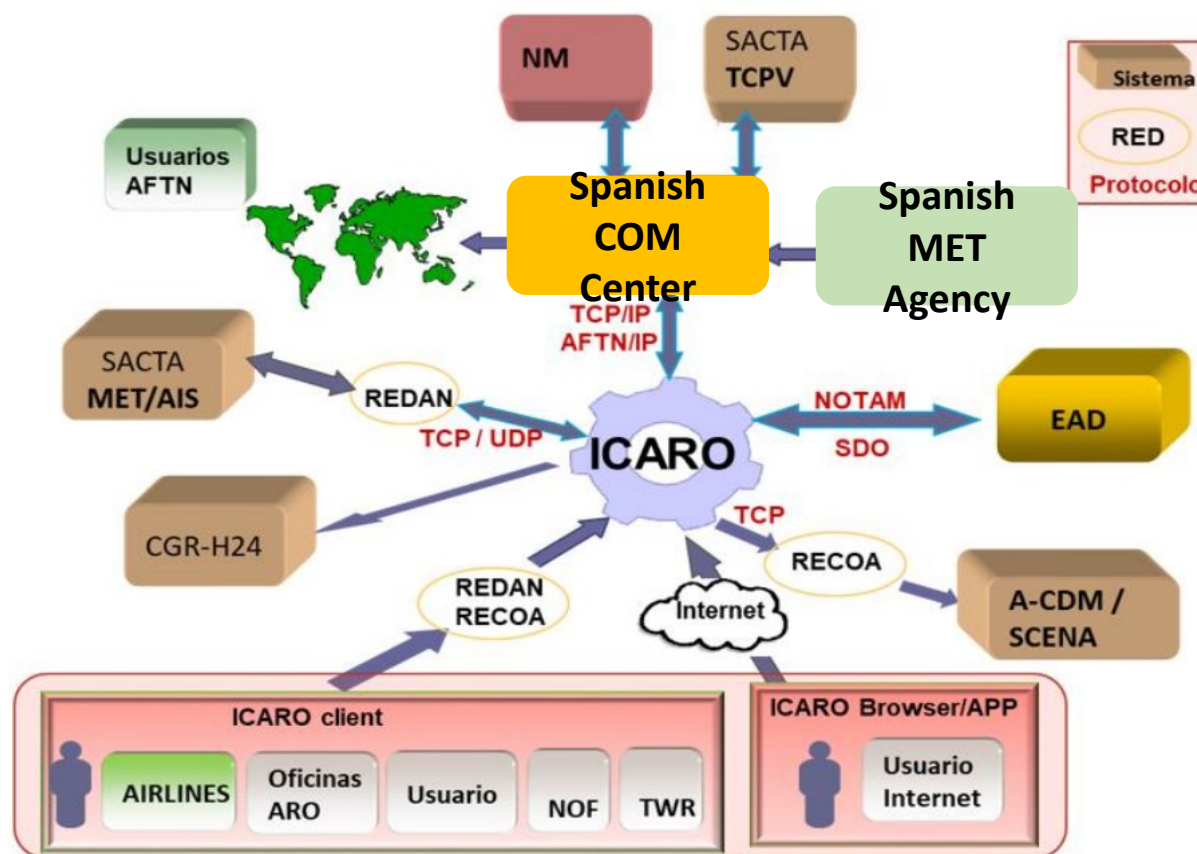
- Use case #3 (By ENAIRE, MET User (SCIMAT))





Rationale!

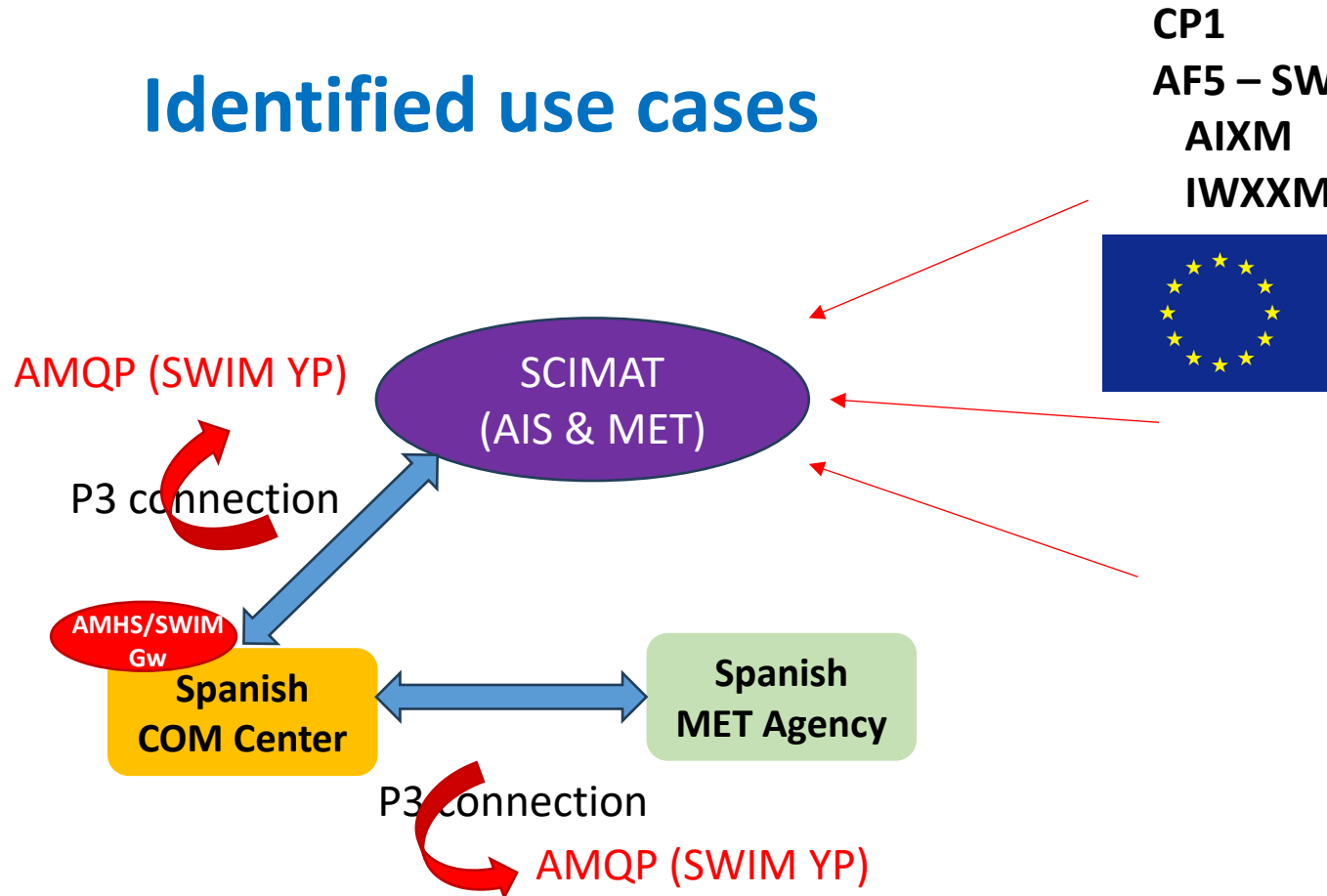
Identified use cases



- ICARO is a system distributing FPL and MET information
- Some of the Spanish users are still AFTN/IP users, e.g., GCCCATS (Canaries contingency FDP System), LESCYPX (Centralized FDP System), etc.)
- These users will receive XML information because of CP1 regulation, however XML doesn't fit within AFTN/IP
- These users will use the AMHS/SWIM Gw rather than implementing a “dual stack” solution (AFTN/IP and AMQP)
- This approach is possible whenever there is no change in the application (NOTAM → DNOTAM)



Identified use cases



- European CP1 Mandate → AF5 SWIM, Deadline Dec 2025
- Progressive migration from legacy technologies (AFTN/IP, AMHS) to SWIM YP (AMQP)
- With the use of the AMHS/SWIM Gw there is no impact on the final users



European strategy towards FF-ICE migration

■ Context and framework

■ CP1 Mandate (EU 2021/116)

■ Implementation of AF5:

- Family 5.1.1. Common infrastructure components (PKI, SWIM Registry)
- Family 5.1.2 SWIM YP technical infrastructure and specifications (Protocols, standards, use of NewPENS, etc.)
- Family 5.1.3 Aeronautical information exchange (ARES, digital NOTAMs, Airspace use plans, ...)
- Family 5.1.4 MET information exchange (MET information for aerodromes, ENR, aids, ...)
- Family 5.1.5 Cooperative network information exchange (Network operations, ATFCM, ...)
- Family 5.1.6 Flight information exchange (FF-ICE/R1, FUM, ...)

31/12/2024

31/12/2025





European strategy towards FF-ICE migration

- Implementation of AF5:
 - Family 5.1.6 Flight information exchange (FF-ICE/R1, FUM, ...)
 - Scope:
 - Related to FF-ICE Release 1 Services:
 - flight plan and routes generation and validation;
 - flight plans, 4D trajectory, flight performance data, flight status;
 - flights lists and detailed flight data;
 - Related to flight update departure information;
 - Flight update messages ('FUM') (Network Manager Business to Business (B2B) service)



31/12/2025



31/12/2025

European strategy towards FF-ICE migration

- FF-ICE Release 1:
 - Different European projects on going to migrate from current ICAO 2012 FPL/ADEXP format to FF-ICE/R1 (eFPL):
 - Make use of available and additional information to better plan and manage ATM operations
 - Integration of relevant flight and flow information in ANSP's ATM systems (FDPS)
 - Better coordination among ANSPs, Airport Operators and the NM to profit from the new information available on eFPL
 - Improvement on operational filing of eFPL for airlines

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